APPLIED MATHEMATICS • GNU OCTAVE & PYTHON

Multivariable Calculus & PDEs in Quantitative Finance

Real-time implied volatility surface reconstruction, numerical root-finding, and gradient vector fields

Beyond Constant Volatility Assumptions

Theoretical Assumption

Classical models assume constant asset volatility across strikes.

Market Reality

Live option markets price crash protection at steep premiums.

Multivariable Solution

Model implied volatility as a smooth 3D surface over strike and maturity.



The Black-Scholes Parabolic PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0$$

Time Decay (Θ)

First partial derivative wrt time.

Convexity (Γ)

Second partial derivative wrt spot price.

Drift Balance (Δ)

Risk-neutral dynamic hedging ratio.

Hybrid Python & GNU Octave Computational Pipeline

STEP 01

gui_app.py

Tkinter interface dispatches async worker thread.

STEP 02

fetch_data.py

Pulls live yfinance chains into 2D meshgrids.

STEP 03

market_data.mat

Binary bridge transfers matrices K, T, S.

STEP 04

iv_surface.m

Octave solves PDE inversion and 3D plots.

Constructing the 2D Coordinate Meshgrid (K, T)

22

Uniform strike nodes

±7%

Moneyness band around spot

5

Active maturity slices

110

Non-linear root equations

```
common_strikes = np.linspace(spot_price * 0.93, spot_price * 1.07, 22)
 K_grid, T_grid = np.meshgrid(common_strikes, maturities_list)
```

Newton-Raphson Volatility Inversion

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{mkt}}{\frac{\partial C_{BS}}{\partial \sigma}}$$

Objective Function: Minimize pricing residual below 1e-6.

Analytical Jacobian: Divide residual by Black-Scholes Vega.

Safeguard Clamping: Bound surface between 5% and 150%.

```
% iv_surface.m: Newton-Raphson loop
for iter = 1:50
  bs_price = S * bs_N_d1 - k_val * exp(-r*t_val) * bs_N_d2;
  diff = bs_price - mkt_price;
  if abs(diff) < 1e-6, break; end
  n_d1 = (1 / sqrt(2*pi)) * exp(-0.5 * bs_d1^2);
  vega = S * sqrt(t_val) * n_d1;
  sigma = sigma - diff / vega;
end
```

Surface Sensitivities via Octave Gradient

```
[dIV_dK, dIV_dT] = gradient(IV);
```

$$\nabla\sigma(K, T) = [\partial\sigma/\partial K, \partial\sigma/\partial T]$$

Strike Sensitivity (∂σ/∂K)

Measures asymmetric crash-risk skew slope.

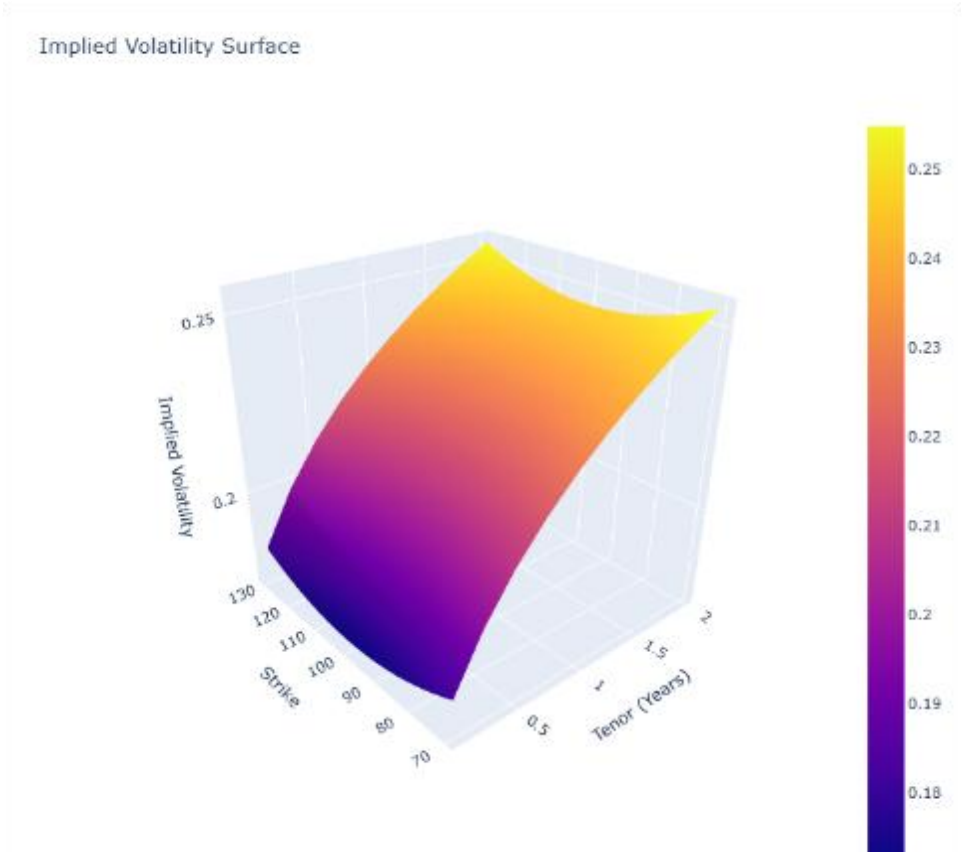
Maturity Sensitivity (∂σ/∂T)

Captures term-structure calendar spread decay.

Finite Differences

Central difference approximation on interior nodes.

Figure 1: Native Octave 3D Surface Topology




```
graphics_toolkit('qt');
 surf(K, T, IV, ...
  'FaceColor', 'interp', ...
  'EdgeColor', 'none');
 camlight; lighting gouraud;
```

Gouraud Shading: Smooth interpolation across 110 grid facets.

Qt Backend: Hardware-accelerated interactive 3D rotation.

2D Orthogonal Slices of the 3D Surface

FIGURE 2 • FIXED MATURITY $T = T_o$

2D Volatility Smile & Skew

plot(K(mid_row, :), IV(mid_row, :), '-ro')

Trace Equation: Evaluates $f(K) = \sigma(K, T_o)$.

Put Skew: Higher implied volatility at low strikes.

Tail Protection: Institutional hedging drives left-tail steepness.

FIGURE 3 • FIXED STRIKE $K = K_o$

2D Volatility Term Structure

plot(T(:, mid_col), IV(:, mid_col), '-bs')

Trace Equation: Evaluates $g(T) = \sigma(K_o, T)$.

Event Spikes: Earnings shocks elevate near-dated maturities.

Mean Reversion: Long-dated volatility stabilizes asymptotically.

Implied Volatility Across Moneyness (K / S)



Automated Output Metrics & Trading Desk Use Cases

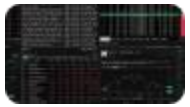
Output Metric (report.txt)	Octave Expression	Desk Application
At-The-Money (ATM) IV	IV(atm_row, atm_col)	Baseline directional option pricing
Surface Extrema (Min / Max)	min(IV(:)) / max(IV(:))	Volatility dispersion bounds
Volatility Skew Spread	otm_put_iv - atm_iv	Tail-risk sentiment gauge
Gradient Vector Field	[dIV_dK, dIV_dT]	Local volatility & exotic hedging

CONCLUSION & LIVE DEMO

Bridging Calculus Theory & Live Markets

PDEs, Newton-Raphson inversion, and 3D gradient fields powered by Python and GNU Octave

Image Sources



https://godelterminal.com/assets/seo/img/hedge-funds-hero-poster.jpg

Source: godelterminal.com



https://miro.medium.com/1*xjR6hNQchbJSW4VkxX0-_Q.png

Source: medium.com